

무결점 우주발사체 구조를 위한 레이저 초음파 선진비파괴평가기술 개발

Development of Laser Ultrasonic Advanced Nondestructive Evaluation Technology for Zero-defect Space Launcher Structure



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UPI System and Multiple Damage Visualization Algorithms

Introduce ultrasonic propagation imaging system which is able to serve automatic NDE.

Ultrasonic Propagation Imaging (UPI) System^[1]

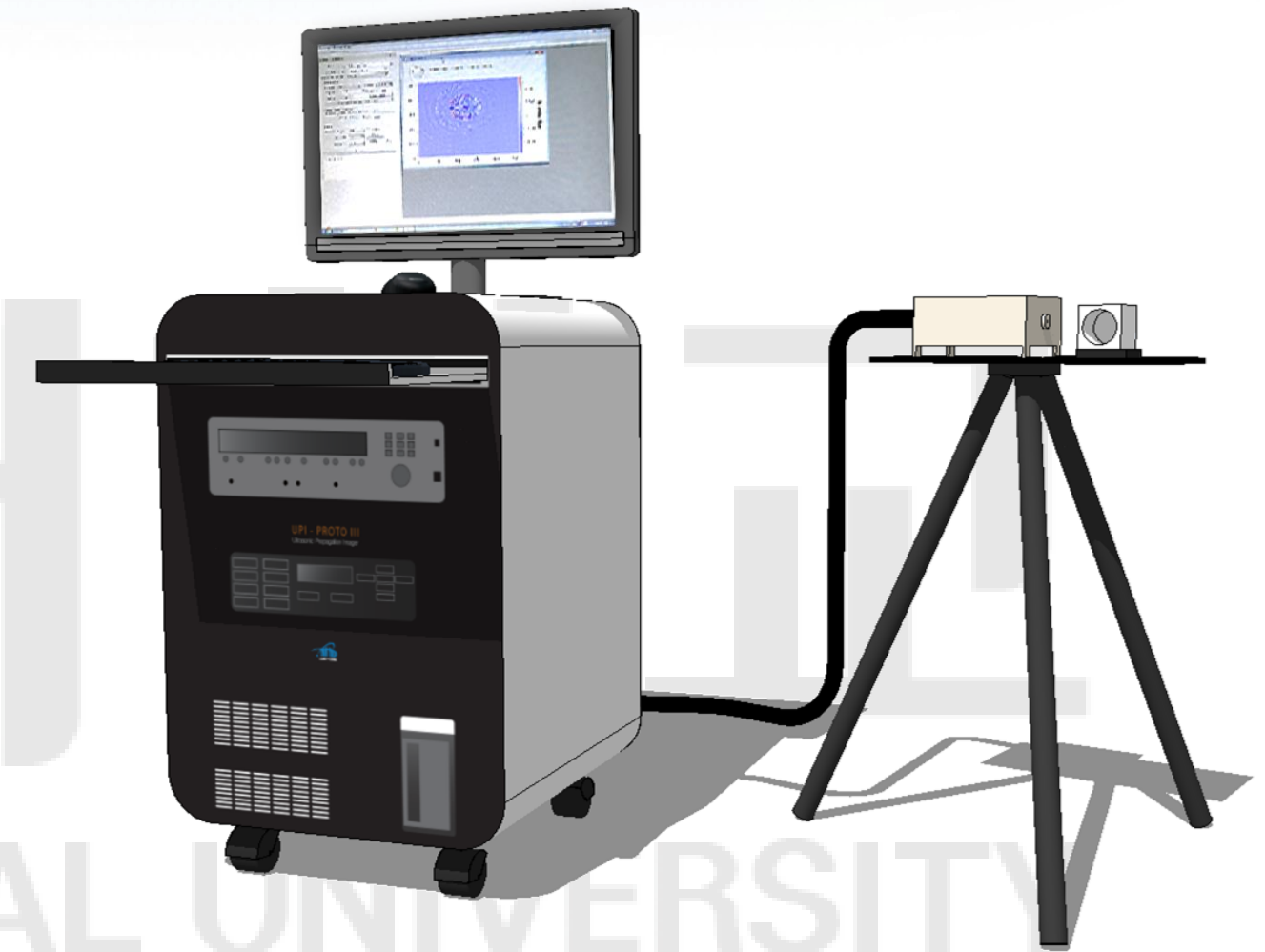
- Use laser excitation as ultrasonic wave generation manner
 - ✓ able to generate ultrasonic wave regardless to shape of target structure
 - ✓ Provide much faster inspection : 50 sec. for 1000 x 1000 mm² inspection area (interval : 1 mm)

Signal processing Algorithms^{[1], [2]}

- The UPI system has been equipped with various signal processing algorithms
 - ✓ UWPI : presents raw ultrasonic wave propagation pattern
 - ✓ MAWS : relatively emphasizes anomalous wave components induced by cracks.
 - ✓ STPM : uses unsimilarities between two adjacent waves induced by cracks.
 - ✓ and 6 more algorithms

^[1] C C Chia, J R Lee, C Y Yoon, C Y Park, J H Kim, Y G Kim, M J Choi, Radome Inspection based on Ultrasonic Frequency Tomography and Ultrasonic Energy Propagation Imaging, Earth & Space 2010

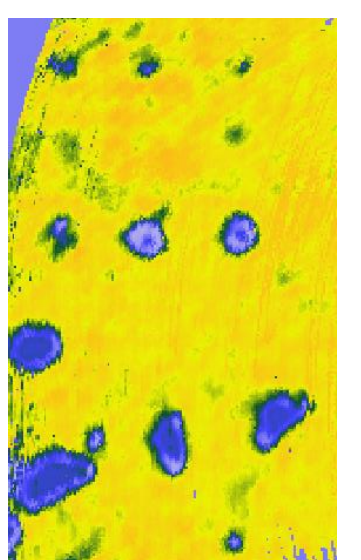
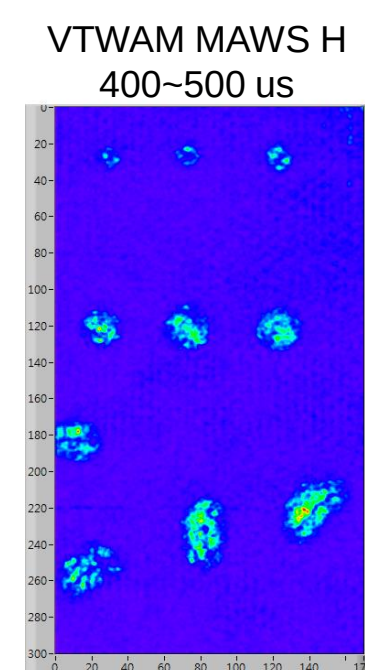
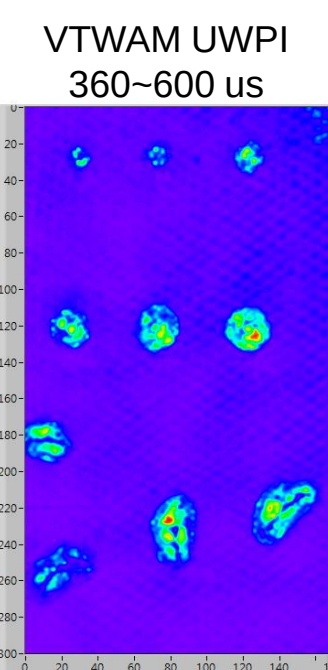
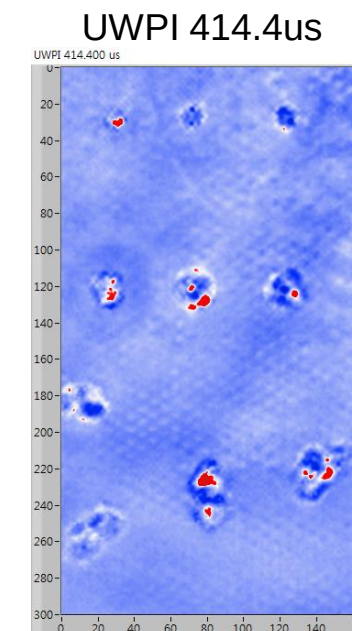
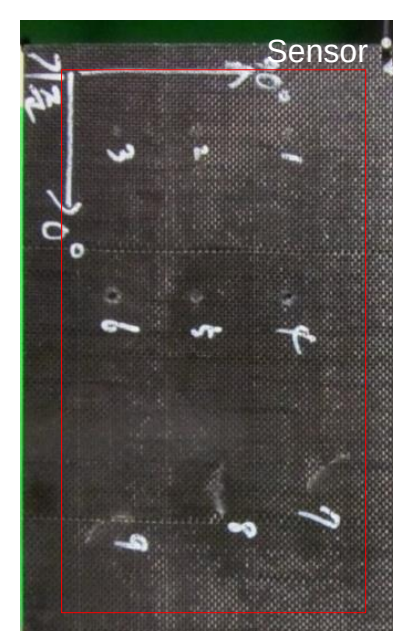
^[2] Y S Choi, H M Jeong, J R Lee*, Laser ultrasonic system for surface crack visualization in dissimilar welds of control rod drive mechanism assembly of nuclear power plant, Shock and Vibration 2014, 296426(10p), March 2014



Damage Visualization Results for Launch Vehicle and Missile Structures

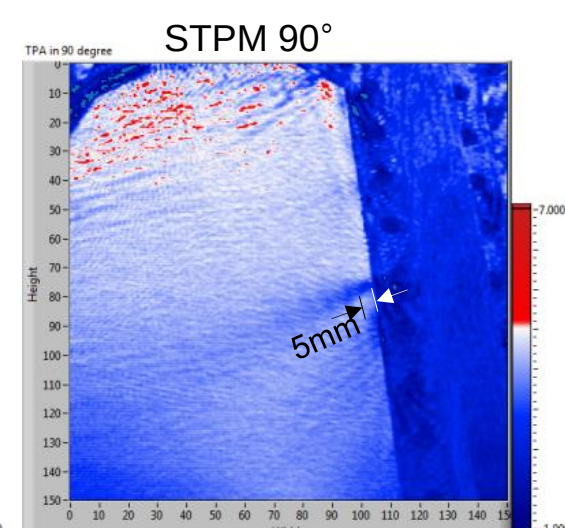
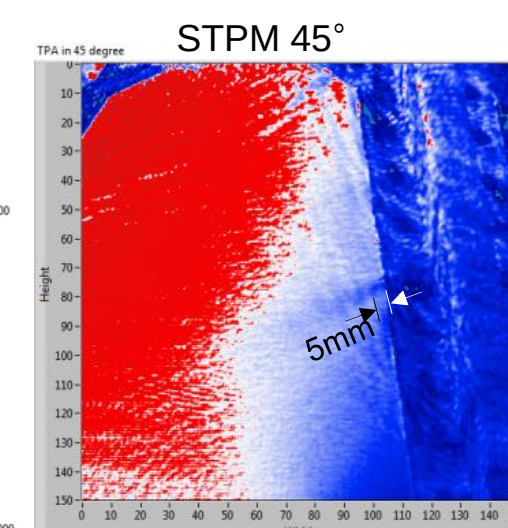
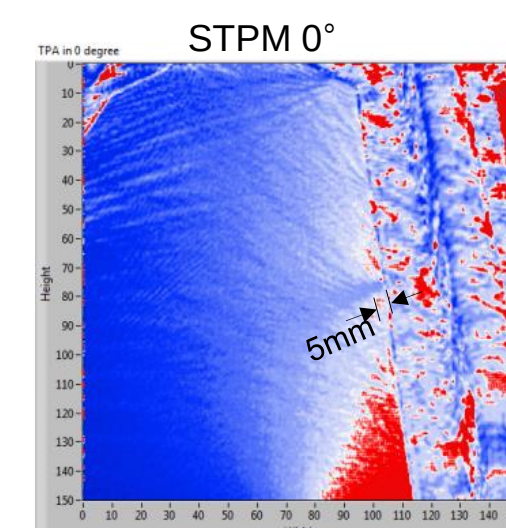
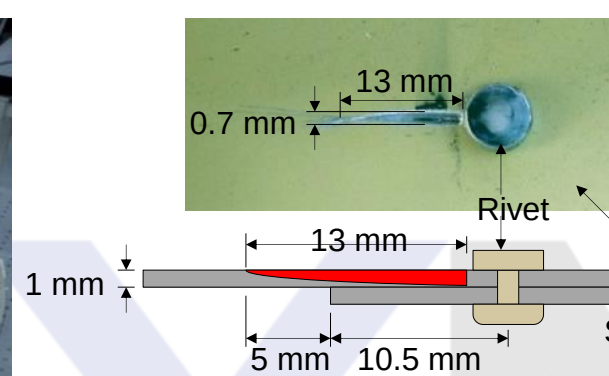
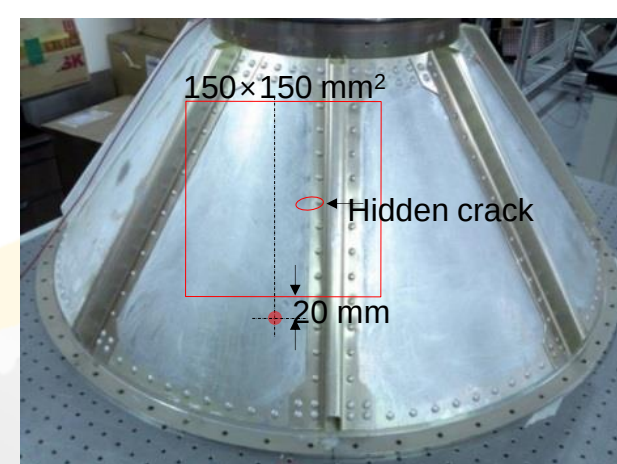
Payload fairing [Aluminum honeycomb core CFRP sandwich panel]

- Impact damages detection



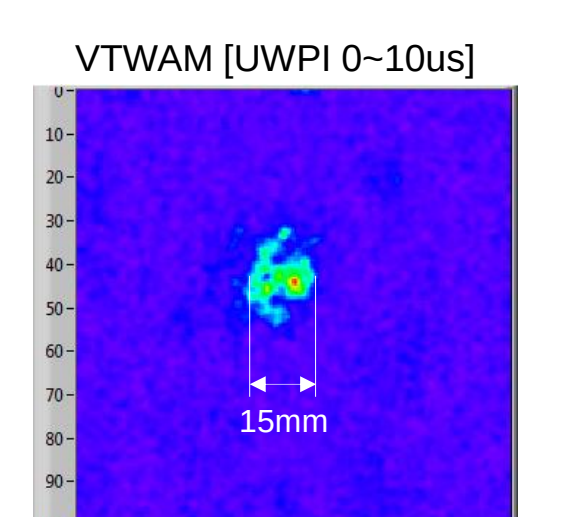
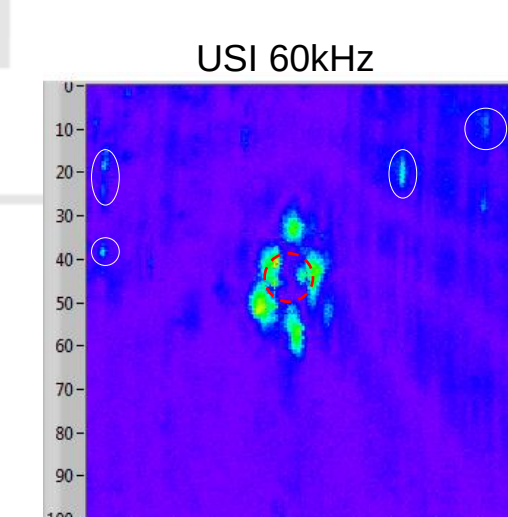
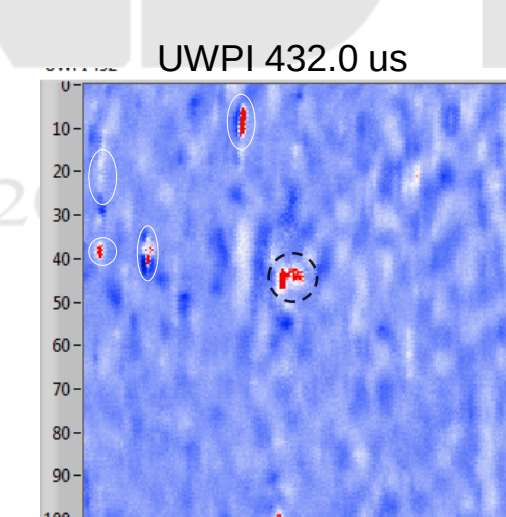
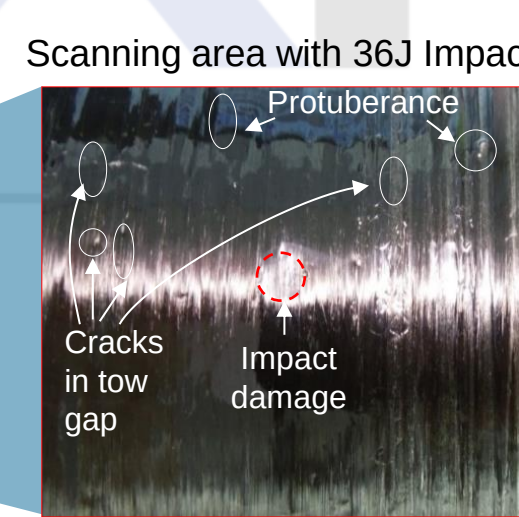
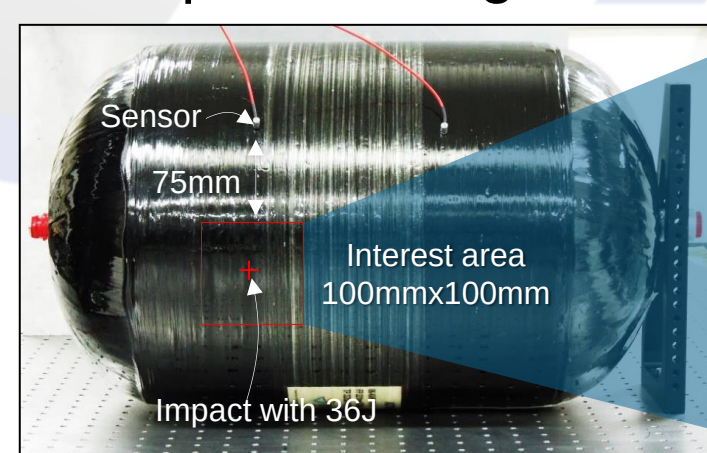
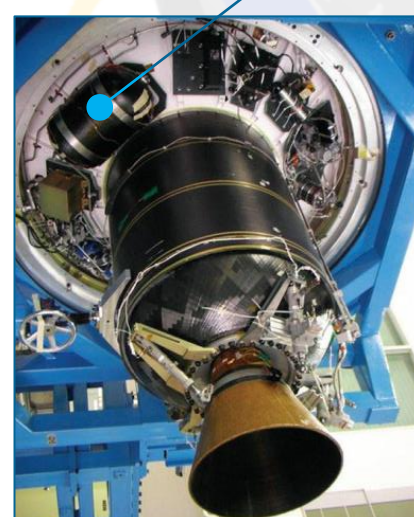
Payload adapter [Aluminum skin and stringer]

- Hidden crack near the rivet



High-pressure tank [Carbon fiber filament wound tank]

- Impact damage

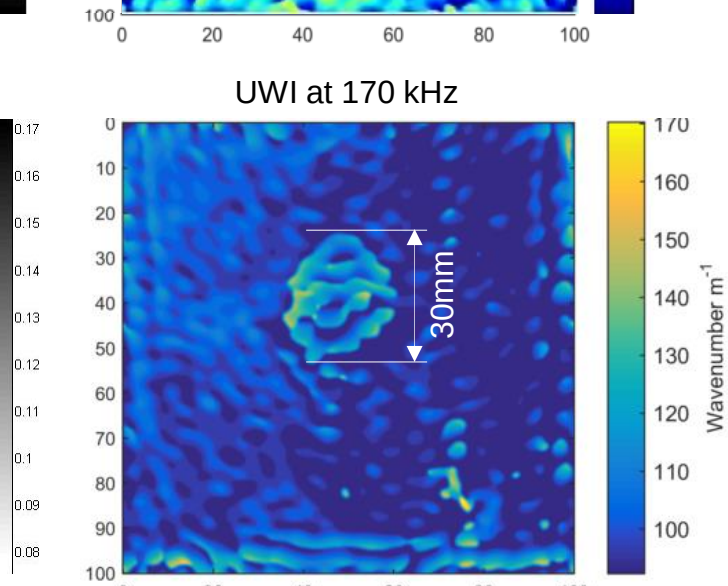
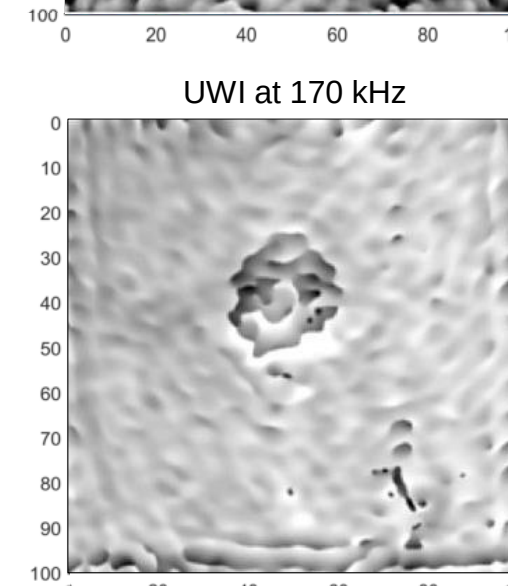
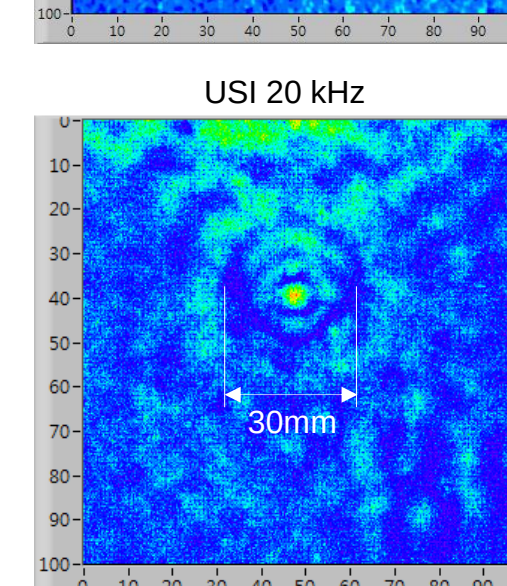
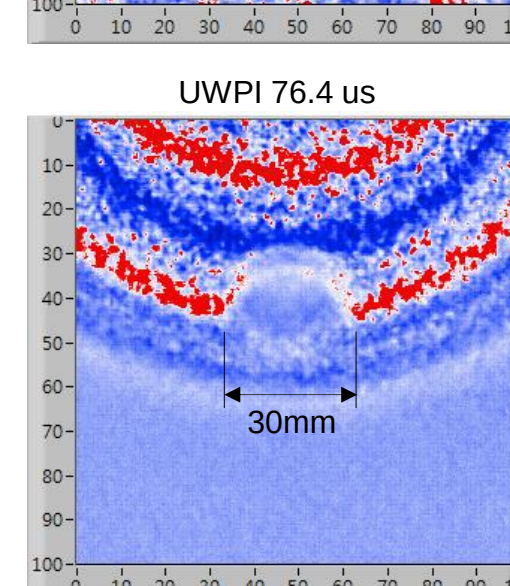
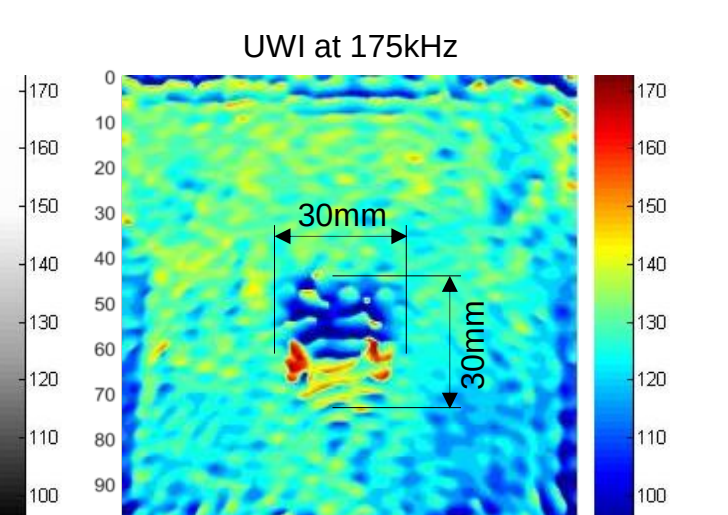
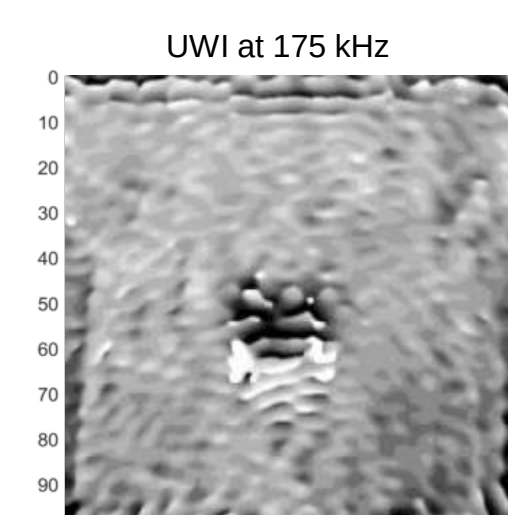
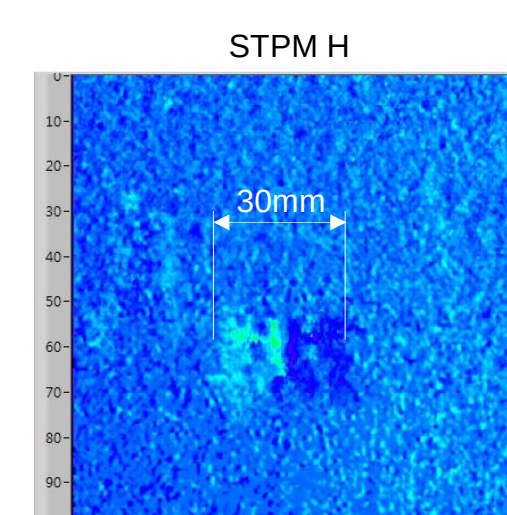
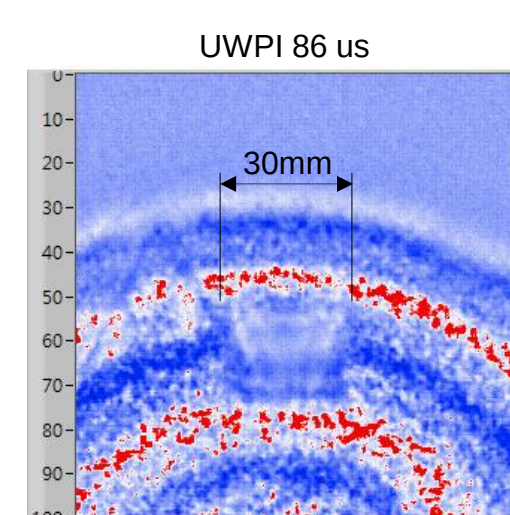
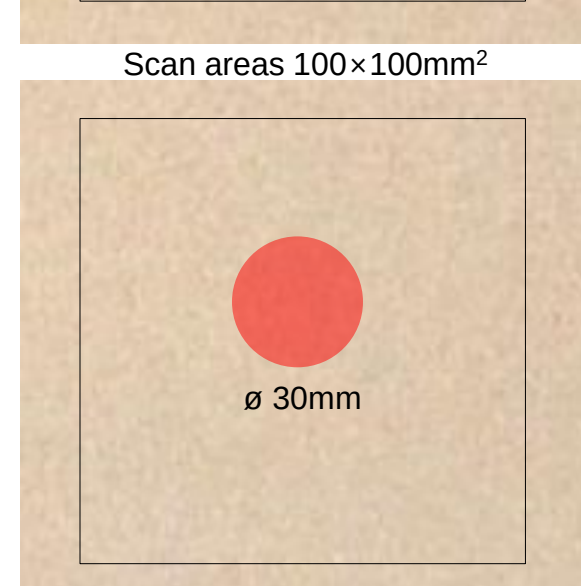
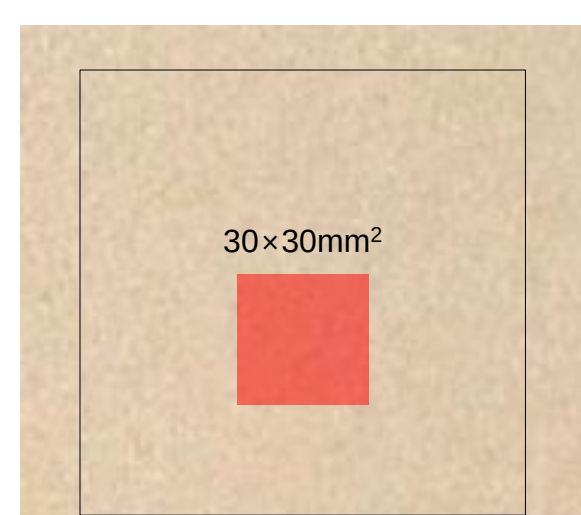
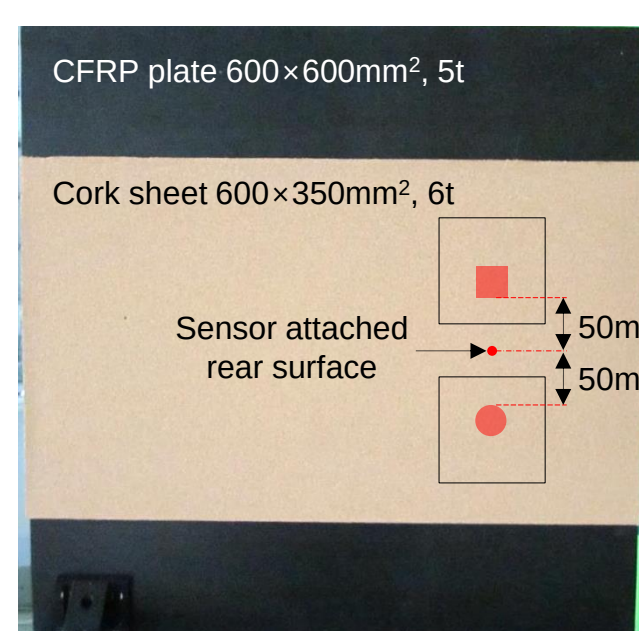


Thermal protection system (TPS) [Cork/CFRP bonded structure]

- Debonding defects



French deterrence force missile



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